SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0702

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Annexure A

Debugging & Optimisation Task

Code of Conduct:

I SHAIK KHAJAH ROSHAN 192511445 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Shaik

20/20

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

<u>Task</u>

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

<u>Task</u>

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	**Next Hop**
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

		and documentation			

Q1, Debugging IPv4 subnet configuration

Given configuration

Parmeter	Configured value
IP Address	192,168,10,65
Subnet Mask	255,255,255,192
CIDR	/26
Default Gateway	192,168,10,1
Intended Subnet	192,168,10,0/26

1, Identify the subnet to which the IP belongs

The subnet mask is:

255,255,255,192 = /26

For /26, the block size is:

256 − 192 = 64

Therefore, the subnets are:

- 192,168,10,0/26
- 192,168,10,64/26
- 192,168,10,128/26
- 192,168,10,192/26

The configured IP is

192,168,10,192/26

Since 65 falls between 64 and 127, it belongs to:

192.168.10.64/26

2, Network Address

For the subnet 192.168.10.64/26:

Network Address = 192.168.10.64

3, Broadcast Address

The next subnet begins at 192.168.10.128.

Therefore, the address immediately before it is:

Broadcast Address = 192.168.10.127

4, Valid Host Range

The network address and broadcast address cannot be assigned to normal hosts.

Therefore:

First usable host : 192.168.10.65

Last usable host : 192.168.10.126

So the valid host range is:

192.168.10.65 – 192.168.10.126

5, Verify the IP Address

Configured IP:

192.168.10.65

Valid range:

192.168.10.65 – 192.168.10.126

Therefore:

192.168.10.65 is a valid host IP address.

6, Verify the Default Gateway

Configured gateway:

192.168.10.1

But the workstation belongs to:

192.168.10.64/26

Valid host range:

192.168.10.65 – 192.168.10.126

The gateway 192.168.10.1 belongs to the 192.168.10.0/26 Subnet, not the workstation's Subnet.

Therefore:

192.168.10.1 is an invalid default gateway for this workstation.

The gateway must be in the same subnet.

7, Identify the configuration errors

There are two important points:

Error 1 – wrong intended Subnet.

The task says the workstation should communicate with systems in:

192.168.10.0/26

But the workstation's IP 192.168.10.65/26 actually belongs to:

192.168.10.64/26

So the workstation is not in the intended subnet.

Error 2 - Wrong Default Gateway

The gate way is:

192.168.10.1

This belongs to:

192.168.10.0/26

While the workstation is in:

192.168.10.64/26

Thus, the gateway and workstation are on different /26 subnets.

8. Optimize / Correct the configuration

There are two possible corrections.

Option A - Keep the intended subnet

192.168.10.0/26

Since the other systems are configured in 192.168.10.0/26 this is the most appropriate correction.

Choose a valid host IP such as:

IP Address: 192.168.10.2

Subnet Mask: 255.255.255.192

Default Gateway 192.168.10.1

So the corrected configuration is:

Setting	correct configuration
IP Address	192.168.10.2
Subnet Mask	255.255.255.192 (/26)
Default gateway	192.168.10.1
Network	192.168.10.0/26

Now both the workstation and gateway belong to the same subnet.

9, Total Number of usable Host Addresses

For /26 :

Number of host bits.

$32 - 26 = 6$

Total addresses :

$2^6 = 64$

Two addresses are reserved:

- Network address
- Broadcast address

Therefore :

Usable hosts $= 64 - 2 = 62$

Final Answer :

Total usable host addresses $= 62$

Q2. IPv6 Address Debugging

Given

Device A: 2001:db8::ff00:42:8329/64

Device B: 2001:db8:0:0:1::1/64

1. Expand both IPv6 addresses

Device A

Compressed:

2001:db8::ff00:42:8329

Expanded:

2001:0db8:0000:0000:0000:ff00:0042:8329/64

Device B

Compressed

2001:db8:0:0:1::1

Expanded:

2001:0db8:0000:0000:0001:0000:0000:0001/64

2. Determine the network prefix

Because the prefix is /64 the first 64 bits = first 4 groups represent the network.

Device A

2001:0db8:0000:0000

0000:ff00:0042:8329

Network prefix:

2001:db8:0:0::/64

Device B

2001:0db8:0000:0000

0001:0000:0000:0001

Network prefix:

2001:db8:0:1::/64

3, Are they in the same subnet?

Compare:

Device A : 2001:db8:0:0::/64

Device B : 2001:db8:0:1::/64

The first four groups are:

- Device A → 2001:db8:0:0
- Device B → 2001:db8:0:0

Therefore:

Yes, both addresses have the same first 64 bits.

Important: The given Device B address 2001:db8:0:0:1::1/64 is not valid IPv6 notation as written. The correct normalized form should have exactly 8 hextets. So the task's Device B address appears to contain an addressing error.

4, Identify the addressing / Prefix mismatch

The likely error is that Device B has an extra 1 in the address.

A corrected Device B address can be :

2001:db8:0:0:1:0:0:1/64

But if the intention is for both devices to be in the same /64 subnet, a clearer address would be :

Device A : 2001:db8:0:0:0:ff00:42:8329/64

Device B : 2001:db8:0:0:0:0:0:1/64

Both belong to :

2001:db8:0:0::/64

5, Optimized IPv6 configuration

Device	IPv6 Address	Prefix
A	2001:db8:0:0:0:ff00:42:8329	/64
B	2001:db8:0:0:0:0:0:1	/64

Both are now in :

2001:db8:0:0::/64

6, Number of available host addresses

For IPv6 /64 :

Host bits :

128 - 64 = 64 bits

number of addresses:

2^{64} = 18,446,744,073,709,551,616

Therefore

Available IPv6 addresses =
18,446,744,073,709,551,616

IPv6 does not use the IPv4-style subtraction of 2 for network and broadcast addresses.

Q3. IPv4 Subnetting and VLSM

Given network:

192.168.1.0/24

Subnets used:

- /26
- /27
- /28

1. /26 Subnet

Mask:

255.255.255.192

Block size:

256 − 192 = 64

First /26

Item	Address
Network	192.168.1.0
First Host	192.168.1.1
Last Host	192.168.1.62
Broadcast	192.168.1.63
Usable Hosts	62

Second /26

Item	Address
Network	192.168.1.64
First Host	192.168.1.65
Last Host	192.168.1.126
Broadcast	192.168.1.127
Usable Hosts	62

2, /27 Subnet

Mask:

255.255.255.224

Block Size:

256-224 = 32

example:

Item	Address
Network	192.168.1.128
First Host	192.168.1.129
Last Host	192.168.1.158
Broadcast	192.168.1.159
Usable Hosts	30

3. /28 Subnet

Mask:

255.255.255.240

Block Size:

256 − 240 = 16

Example:

Item	Address
Network	192.168.1.160
First Host	192.168.1.161
Last Host	192.168.1.174
Broadcast	192.168.1.175
Usable Hosts	14

4. Number of usable addresses

Prefix	Total Addresses	Usable Addresses
/26	64	62
/27	32	30
/28	16	14

5. Identify inefficient allocation

A /26 provides 62 usable addresses.

If a department needs only for example, 10-15 hosts, allocating a /26 wasters many addresses.

similarly, a /27 gives 30 usable addresses, which may be excessive for a small department.

Therefore:
Large subnets allocated to departments with small requirements are inefficient.

6, VLSM Optimization

VLSM = Variable Length Subnet Masking
Instead of giving every department the same-size subnet, assign subnet sizes according to actual requirements.

Example:

Department Requirement	Recommended Subnet	Usable Hosts
Large department	/26	62
Medium department	/27	30
Small department	/28	14

This reduces unused IP addresses

Conclusion

VLSM is better because:

- It allocates addresses according to department size.
- It reduces IP wastage.
- It provides better address utilization.
- It allows different subnet sizes within the same /24 network.

Q.4. IPv6 Address Debugging

Given

Device A:

2001:db8::ff00:42:8329/64

Device B:

2001:db8:0:011::1/64

Expanded Device A

2001:db8:0000:0000:0000:ff00:0042:8329/64

Expanded Device B

As written, the address has an invalid group structure.

The intended form is likely:

2001:0db8:0000:0000:0001:0000:0000:0001/64

Network Prefix

Device A:

2001:db8:0:0::/64

Device B:

2001:db8:0:0::/64

assuming the intended 8-hextet form above.

Correct configuration

For a clean same-subnet configuration:

Device A: 2001:db8:0:0:0:ff00:42:8329/64

Device B: 2001:db8:0:0:0:0:0:1/64

Network :

2001:db8:0:0::/64

Available addresses:

$2^{64} = 18,446,744,073,709,551,616.$

Q.5. Routing Table Debugging

Given Routing Table

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Destination :

192.168.2.100

1. Find the matching routing entry

Destination IP :

192.168.2.100

Check the routes :

Route 1

192.168.1.0/24

Does not match because destination is 192.168.2.100.

Route 2

192.168.2.0/24

Matches.

The range is:

192.168.2.1 – 192.168.2.254

Therefore:

192.168.2.100 belongs to 192.168.2.0/24

2, Longest Prefix Match

The router select the most specific matching route.

Matching route:

192.168.2.0/24

Next hop:

10.0.0.3

Therefore:

Packet will be forwarded to 10.0.0.3.

3, Verify destination Subnet

Subnet:

192.168.2.0/24

Network address:

192.168.2.0

Broadcast:

192.168.2.255

valid hosts:

192.168.2.1 – 192.168.2.254

Since:

192.168.2.100

4, Identify routing configuration errors

The routing table itself contains the correct route for 192.168.2.100.

Therefore:

There is no obvious routing-table error for this destination.

If packets are still not reaching the destination check:

1, whether 10.0.0.3 is reachable.

2, whether interfaces are up.

3, whether return routing is correctly configured.

5, Optimize the routing table

The table alread uses specific /24 routes plus a default route.

An optimized table is:

Destination	Next Hop	purpose
192.168.1.0/24	10.0.0.2	Specific route
192.168.2.0/24	10.0.0.3	Specific route
0.0.0.0/0	10.0.0.1	Default route

This provides:

- specific routing for known networks.
- Default routing for unknown networks.
- Efficient packet forwarding through longest-prefix matching.

6, what happens if no matching ~~to~~ destination exists?

If the destination does not match:

- 192.168.1.0/24
- 192.168.2.0/24

the router uses the:

Default Route : 0.0.0.0/0

Next hop:

10.0.0.1

For example, if the destination is :

192.168.5.100

there is no specific route, so :

Packet → 10.0.0.1